

A Question examples

2-hop	Question: Which book is shown during her train ride and then later shown again when she reacts on the train? Answer: The Vegetarian. Golden-Clip: [14, 25] Reasoning-Chain: Step 1: Slice 14 shows the book titled ‘The Vegetarian’ being read on transit. Step 2: Slice 25 shows her on a train with a red book and the text about gasping on the train. Conclusion: The repeated book is ‘The Vegetarian.’ Category: Visual_Tracking Hop-level: 2-Hop Visual_proof: In Clip 1, the text overlay explicitly states..... as she reacts to it. Logic_check_reasoning: Step 1: Slice 14 explicitly shows a person on a train it again during her reaction. Video-id: -uvMrMcN0eA
3-hop	Question: Across the Italy-focused maps, which trio of cities repeatedly appears as labeled key points (including one that later becomes central to the siege narrative)? Answer: Rome, Ravenna, and Naples. Golden-Clip: [43, 50, 72] Reasoning-Chain: Step 1: Slice 43 labels Rome, Ravenna, and Naples on a Kingdom of Italy map. Step 2: Slice 50 again labels Rome, Ravenna, and Naples among marked cities. Step 3: Slice 72 repeats Rome and Naples prominently and includes Ravenna in the same political landscape. Conclusion: The recurring trio is Rome, Ravenna, and Naples. Category: Global_Summary Hop-level: 3-Hop Visual_proof: All three clips show a map of the Kingdom of Italy..... supporting the narrative context. Logic_check_reasoning: Step 1: Slice 43 lists the citiesshows the trio as labeled cities on Italy-focused maps. Video-id: -7wwfGJXEZg
4-hop	Question: How does her dessert-prep storyline progress from choosing a fruit at the store to a final plated result that includes more than one kind of fruit? Answer: She selects strawberries at the store, then later dips strawberries (and includes grapes too) in chocolate, ending with a final results plate containing strawberries and grapes. Golden-Clip: [35, 54, 59, 60] Reasoning-Chain: Step 1: Slice 35 shows her choosing strawberries while shopping. Step 2: Slice 54 shows grapes being added and the idea of covered fruits. Step 3: Slice 59 shows dipping strawberries into melted chocolate. Step 4: Slice 60 shows ‘final results’ with strawberries and grapes on a plate. Conclusion: Shopping leads to multi-fruit chocolate prep and a finished mixed-fruit plate. Category: Causal_Inference Hop-level: 4-Hop Visual_proof: Clip 1 shows her selecting strawberries..... chocolate-covered strawberries and grapes. Logic_check_reasoning: Step 1: Slice 35 shows selecting/purchasing strawberries..... to final plated outcome. Video-id: -uvMrMcN0eA

Figure 3: Data examples of different hop. Each block lists the question, answer, golden clips, reasoning chain, category, hop-level, video-id and verification fields.

B The Agentic Construction Pipeline

Our data construction employs a seven-stage “Coarse-to-Fine” pipeline. The adversarial filtration successfully reduced the dataset size from 11,612 raw generations to 3,159 high-quality, retrieval-necessary questions.

Stage 1: Generation. We employ GPT-5.2 as a Generator Agent to mine latent reasoning chains from the dense video captions. By adopting a specific persona, the agent is instructed to identify non-contiguous events that share logical connections and synthesize multi-hop question-answer pairs. This phase prioritizes recall

over precision, generating a diverse pool of 11,612 raw candidates covering various reasoning types.

Stage 2: Rule-based Filtration. Following generation, we apply a strict **Rule-based Filtration** focused on **Temporal Discontinuity**. A defining characteristic of valid multi-hop retrieval is that evidence segments must be scattered across different temporal moments. We discard candidates where the retrieved evidence clips share temporal overlap ($\text{IoU} > 0$), as these effectively degenerate into single-segment reasoning tasks and violate the fundamental multi-hop constraint. This stage retained 11,423 candidates.

Stage 3: Semantic Leakage Removing (The Tautology Filter). A common failure mode in synthetic QA generation is *Answer Leakage*, where the question inadvertently contains the information required to answer it. We deploy a strict Auditor Agent (powered by GPT-5) to identify and discard questions where the answer can be fully inferred or explicitly stated within the question text itself, independent of external context. This rigorous check removed 33.7% of the data, highlighting the prevalence of tautologies in raw LLM outputs.

Stage 4: Logical Solvability Verification. A Verifier Agent (powered by GPT-5) conducts a Chain-of-Thought (CoT) check to identify internal semantic contradictions. It discards queries where the reasoning chain is broken or relies on hallucinations not present in the source captions, retaining 6,784 robust queries.

Stage 5: Adversarial Necessity Check ($N - 1$ Ablation). **This is the core contribution to retrieval rigor.** To eliminate shortcut learning, we implement an *Adversarial Ablation Protocol*. For a question requiring k evidence slices, we mask exactly one slice s_i and challenge a Verifier Agent (GPT-5) to answer using only the remaining context. A question is deemed valid **if and only if** the agent returns “INSUFFICIENT” for all k ablation tests.

This stage rejected 46.0% of the remaining candidates, proving that nearly half of logically valid “multi-hop” questions were not actually retrieval-necessary.

Stage 6: Visual Grounding & Refinement. Since textual captions are lossy compressions of reality, we deploy a *Visual Agent* powered by Qwen3-VL-235B to “watch” the raw video clips corresponding to the generated timestamps. The agent verifies visual consistency (e.g., color, count, action) and generates a refined answer if discrepancies are found.

Stage 7: Final Human Audit. To guarantee benchmark integrity, we implemented a rigorous human-in-the-loop verification protocol acting as the final quality gate.

Annotator Qualification. We recruited a panel of **10 postgraduate researchers** specializing in Computer Vision and Multi-modal Learning. Before the formal audit, all annotators underwent a qualification phase, requiring them to pass a screening test with 100 control samples (50 valid, 50 flawed) with an accuracy threshold of 95% to ensure alignment with our rigorous standards. To prevent cognitive fatigue and ensure high vigilance, we randomly partitioned the 3,392 candidates among the experts, resulting in a manageable workload of **approximately 300 samples per auditor**.

The Adversarial Review Protocol. Unlike passive annotation, experts operated under an “**Adversarial Falsification**” mandate. They were instructed to aggressively search for flaws rather than verify correctness. Using a custom verification interface, auditors watched the raw video segments referenced by the generated timestamps and assessed each QA pair against a strict **Three-Point Rejection Rubric**:

- (1) *Visual Hallucination*: The reasoning relies on visual details not present or ambiguous in the raw pixel data (e.g., misidentifying a blurry object).
- (2) *Logic Loophole*: The reasoning chain contains non-sequiturs or requires external knowledge outside the video context.

- (3) *Retrieval Unnecessity*: The question is theoretically solvable via language priors or a single frame, failing the strict multi-hop requirement.

During this exhaustive audit of all 3,392 machine-verified samples, experts **rejected 233 instances (6.9%)** containing residual ambiguity and manually **modified 4 instances (<0.1%)** to refine linguistic precision. The fact that over 93% of the candidates passed this stringent review without rejection provides compelling evidence for the efficacy of our automated pipeline, confirming that the adversarial filters (Stages 3–6) successfully maintained high data purity.

To strictly quantify the final quality, we conducted a post-audit verification by **randomly sampling 100 QA pairs** for a blind review. The inspection revealed that **100% of these sampled instances were error-free**. This perfect validation rate, combined with the high acceptance rate of the full audit, establishes LONGVID-SEARCH as a high-fidelity benchmark.

C Baseline Agent Workflow

Baseline agent. The baseline agent follows an iterative *plan–retrieve–read–reason* loop. At each step, it formulates a textual query based on the current hypothesis and partial evidence, retrieves candidate clips via `Search_Clips_In_Video`, reads their captions using `Get_Clip_Detail`, and decides whether further retrieval is needed. Once sufficient evidence is collected, the agent outputs `FINAL_ANSWER` with the selected evidence clip IDs. Under this fixed interface, end-to-end performance largely reflects the agent’s ability to generate effective queries and conduct multi-step evidence acquisition.

E Human Evaluation Details and Ethical Considerations

Annotator Recruitment. Human evaluators were recruited from adult participants fluent in English. All annotators were informed of the purpose of the study and participated voluntarily. To ensure the reliability of human judgments, annotators were required to have prior experience in evidence-based QA or text evaluation tasks and to pass a screening test. The screening test assesses: (i) the ability to follow hop-wise evidence requirements, (ii) the ability to distinguish grounded answers from speculation/hallucination, and (iii) the correct handling of insufficient cases (answerable vs. unanswerable given evidence).

Annotation Procedure and Criteria. Annotators were presented with a question, the model prediction, the benchmark-provided reasoning chain with hop-wise key points, and the corresponding evidence clips (IDs and captions/frames). Annotators were asked to assign a binary correctness label following the rubric in Table 7. A prediction is labeled as Correct (Score 1) only if it answers the question correctly *and* satisfies *all* hop-wise evidence requirements supported by the evidence clips; otherwise it is labeled as Incorrect (Score 0), including cases where the model outputs insufficient despite the answer being available from evidence. Detailed guidelines and illustrative examples were provided in advance to reduce ambiguity.

Verification Set Construction. We verify two complementary subsets to validate the reliability of LLM-judge labels: (1) *Disagreement set*: all instances where the three LLM judges (GPT-5, Gemini 3

D Retrieval cost(tool-invocation count) by model, category, and hop level

Table 6: Average retrieval cost by model, category, and hop level.

Agent Backbone	State_Mutation			Causal_Inference			Global_Summary			Visual_Tracking		
	2-hop	3-hop	4-hop	2-hop	3-hop	4-hop	2-hop	3-hop	4-hop	2-hop	3-hop	4-hop
Close-Sourced LLMs												
GPT-5	9.26	9.69	10.13	9.40	10.02	11.02	9.17	9.56	10.50	9.29	10.27	10.23
Gemini 3 Pro	7.35	7.16	7.29	7.21	7.33	7.62	7.37	7.39	7.52	7.29	7.76	7.98
GPT-4o	8.41	8.70	8.81	8.44	8.83	8.99	8.11	8.60	9.22	8.37	8.82	8.82
GPT-4-mini	6.33	6.32	6.37	6.45	6.43	6.72	6.34	6.56	6.59	6.40	6.54	6.43
Open-Sourced LLMs												
Qwen3-VL-32B	8.32	8.52	8.77	8.53	8.89	9.49	8.24	8.57	8.86	8.18	8.66	8.84
Qwen3-VL-8B	8.02	8.05	8.64	8.20	8.33	8.75	7.74	8.29	8.67	7.79	8.17	8.44
Qwen2.5-VL-72B	8.68	9.01	9.32	8.50	9.14	9.74	8.27	8.77	9.58	8.53	9.21	9.48
Qwen2.5-VL-7B	7.19	7.28	7.29	7.17	7.24	7.36	7.16	7.31	7.25	7.16	7.10	7.74
Qwen2.5-7B	7.99	8.20	7.99	8.18	8.03	8.16	8.00	8.08	8.02	7.85	8.36	8.34
Llama-3-8B	7.85	8.03	8.87	7.88	8.39	8.74	7.69	7.96	8.11	7.52	8.50	8.28

Table 7: Human evaluation criteria for the multi-hop question cross three dimensions.

Criterion	Score 1 (Correct)	Score 0 (Incorrect)
Answer correctness	All required entities/values are correct and unambiguous.	Any required detail is wrong/missing, or insufficient despite available evidence.
Hop-wise evidence	Satisfies <i>all</i> hop key points in the reasoning chain; supported by evidence clips.	Misses any hop key point, or uses unsupported speculation/hallucination.
Extra details	Extra details are allowed if they do not contradict evidence or change the answer.	Adds contradictory/fabricated details or changes the answer beyond evidence.

Pro, and GPT-4o) do not reach a majority agreement ($N_{\text{disagree}} \approx 200$ in our experiments), which are adjudicated by domain experts; (2) *Agreement audit set*: an additional 400 instances sampled from the majority-agreed pool for random auditing. To mitigate coverage bias, the audit set is selected in a stratified manner across categories and hop levels. Across all evaluated agent backbones, this protocol yields 6172, human-LLM label comparisons.

Human-LLM Judge Consistency. We compare the final human labels against the LLM majority-vote labels on all verified instances. As shown in Table 5, the overall mismatch rate is 0.87% (54/6172), and per-backbone mismatch rates remain around 1% or lower, indicating that the LLM majority-vote evaluation is stable under our hop-aware rubric.

Compensation. Annotators were compensated at a fixed rate of \$15 per hour. The compensation was independent of annotators’ ratings to avoid incentive bias, and no performance-based or outcome-dependent rewards were provided.

Ethical Considerations. The human evaluation process did not involve the collection of any personally identifiable information. All evaluation content was anonymized and contained no sensitive personal data. Annotators were informed that the evaluated responses were model-generated and were instructed to focus solely on evidence-grounded correctness. Given the non-invasive nature

of the task and the absence of personal data collection, this study does not raise significant ethical concerns and does not require institutional review board approval, consistent with prior work in NLP and multimodal evaluation.